BANGLADESH UNIVERSITY OF PROFESSIONALS

Faculty of Science & Technology (FST)

Department of Computer Science and Engineering

Course Name : Software Engineering Laboratory
Course Code: CSE-3206
Group no : 06

Project Name: Smart Expense Tracker

Group Member :
1. Al Sadique Nuhin - 23524202061
2.Md Junayet Hossain Mohit - 23524202065
3.Abdullah Mahmud Yamin - 23524202085
4.Mashrafi Elahi - 23524202137

Submission Date: 02 April 2026

Design of a Use Case Diagram for a Software System

Smart Expense Tracker — Use Case Diagram & User Stories

1. Objective(s)

- To understand the concept of use case diagrams in UML.
- To identify actors and use cases of the Smart Expense Tracker system.
- To represent system functionality graphically.
- To define user stories for each use case.

2. Problem Analysis

The Smart Expense Tracker is a web-based personal finance management application. Users need an efficient way to record, categorize, and analyze their daily expenses. The system must support registration, secure login, expense management (add, edit, delete), categorization, expense history viewing, and automated report generation. An admin may also manage user accounts and monitor system activity.

3. Use Case Diagram

A Use Case Diagram is a type of UML (Unified Modeling Language) diagram that represents the functional requirements of a system. It shows the interaction between actors (users or external systems) and the system.

3.1 Key Components

Actor: Represents a user or external entity interacting with the system. In this system: User (General User) and Admin.

Use Case: Represents a function or service provided by the system. Represented by ovals; they denote system functionalities.

System Boundary: Defines the scope of the Smart Expense Tracker. All use cases within the boundary belong to the system.

Relationships: Association (solid line), <> (mandatory inclusion), <> (optional extension).

3.2 Use Case Diagram of Smart Expense Tracker

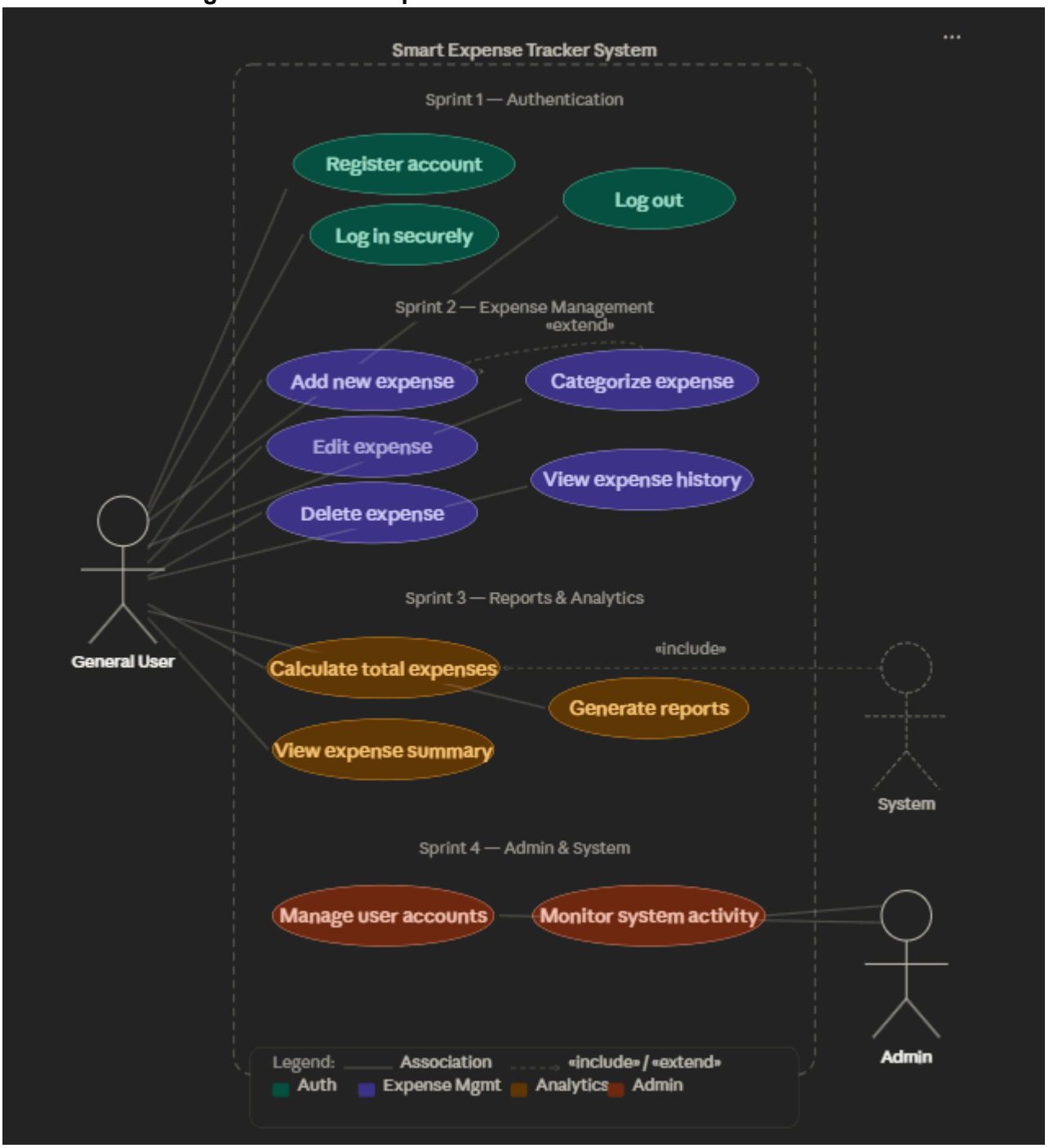

Figure 1: Use Case Diagram of Smart Expense Tracker

4. Methodology

- Identify the system scope — Smart Expense Tracker (web-based expense management).
- Identify all actors interacting with the system (User, Admin).
- Identify all possible use cases from functional requirements.
- Establish relationships between actors and use cases (associations, include, extend).
- Draw the diagram with proper UML notation.

5. Actors and Use Cases

Actors:

- **User (General User)** — The primary actor who registers, logs in, and manages personal expenses.
- **Admin** — An optional actor who manages user accounts and monitors overall system performance.

Use Cases:

Use Case ID	Use Case Name	Actor(s)	Relationship
UC-01	Register Account	User	Association
UC-02	Login	User	Association (<<include>> by UC-03 to UC-07)
UC-03	Add Expense	User	Association, <<include>> Login, Categorize
UC-04	Edit Expense	User	Association, <<include>> Login
UC-05	Delete Expense	User	Association, <<include>> Login
UC-06	Categorize Expense	User	<<include>> from Add Expense
UC-07	View Expense History	User	Association, <<include>> Login
UC-08	Calculate Total Expenses	System	<<include>> from Add Expense
UC-09	Generate Reports	User	<<include>> from View History
UC-10	Manage User Accounts	Admin	Association
UC-11	View System Overview	Admin	Association

6. User Stories

The following user stories define the expected behavior of the Smart Expense Tracker from the perspective of each actor:

As a new user, I want to register an account using my name, email, and password so that I can securely access the system and manage my personal expenses. Once registered, as a user, I want to log in with my credentials so that I can access my account and use all features of the application.

As a user, I want to add a new expense by entering details such as amount, date, category, and description so that I can keep track of my daily spending. I also want the ability to edit any existing expense record so that I can correct mistakes if I entered something incorrectly. Additionally, I want to delete any expense that is no longer needed so that my expense history remains clean and relevant.

As a user, I want to categorize my expenses (such as Food, Transport, Bills, and Shopping) so that I can better understand where my money is going. I also want to view my complete expense history so that I can review all my past transactions in one place.

Furthermore, I want the system to automatically calculate my total expenses so that I can easily see how much I have spent over a certain period. I also want to generate reports and summaries of my expenses so that I can analyze my financial behavior and make better decisions.

As an admin, I want to manage user accounts so that I can add, remove, or deactivate users when necessary. Additionally, I want to view an overall system overview, including performance and user activity, so that I can ensure the application is running efficiently and securely.